We’re worrying about the wrong thing:

the matched guise technique and social speech perception

Kyler Laycock
laycock.21@osu.edu

Kevin B. McGowan
kbmcgowan@uky.edu



MidPhon 28
Purdue University
21 October 2023



Matched Guise

Goal: Classically, sociolinguists use Matched Guise to determine the influence of indexical cues on language attitudes (Kircher 2015; Campbell-Kibler 2009; Lambert et al. 1960)

Task:

- a linguistic signal performed by a single talker
- two or more invoked social categories
- elicit an attitudinal response

Example: Matched Guise

Crosslinguistic Perceptions of /s/

Boyd, Fruehwald, and Hall-Lew (2021)

- Recordings of four cisgender males: first language English (England, Essex), French (France, Lyon), German (Germany, Düsseldorf), and Estonian (Estonia, Püünsi)
- Stimuli created by splicing natural fronted, neutral, and backed productions of /s/ produced by Boyd into the original recordings

Boyd, Fruehwald, and Hall-Lew (2021)

Crosslinguistic Perceptions of /s/

- Resulting auditory stimuli
    - [s+] (fronted, high CoG)
    - [s] (neutral), and
    - [s-] (backed, low CoG)
- Spliced into in each of the four languages

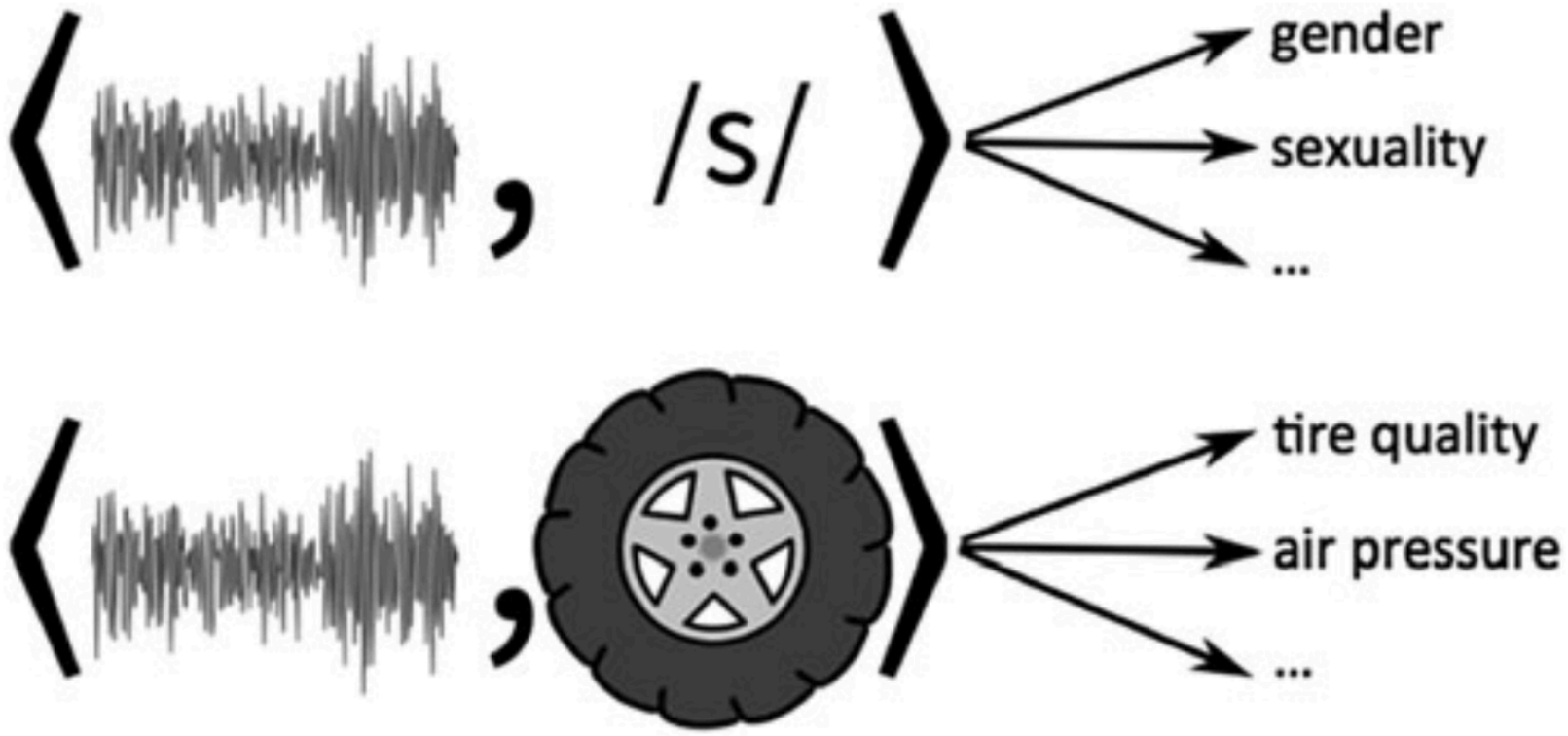


Fig. 1 from Boyd, Fruehwald, and Hall-Lew (2021)

Boyd, Fruehwald, and Hall-Lew (2021)

Crosslinguistic Perceptions of /s/

- First-language English, French, and German speakers were recruited as participants
- Each participant heard all guise conditions (auditory stimuli) and provided ratings on six scales:
- Straight/Gay, Educated/Uneducated, Lazy/Hardworking, Friendly/Mean, Masculine/Effeminate, Natural/Synthetic

Boyd, Fruehwald, and Hall-Lew (2021)

Crosslinguistic Perceptions of /s/

Relevant Findings:

- English listeners associate fronted [s+] and Gayness regardless of language, regardless of knowledge of language
- French and German listeners do *not* associate fronted [s+] with Gayness for any language (including their own) even though this feature is present in the natural speech of gay men in both French and German (Boyd 2018a, 2018b)

Speech Perception: Inverted Matched Guise

Goal: Inverted Matched Guise (McGowan 2015) holds the linguistic signal constant and manipulates social expectations in a speech perception task (cf Drager 2013, “identification task”)

Task:

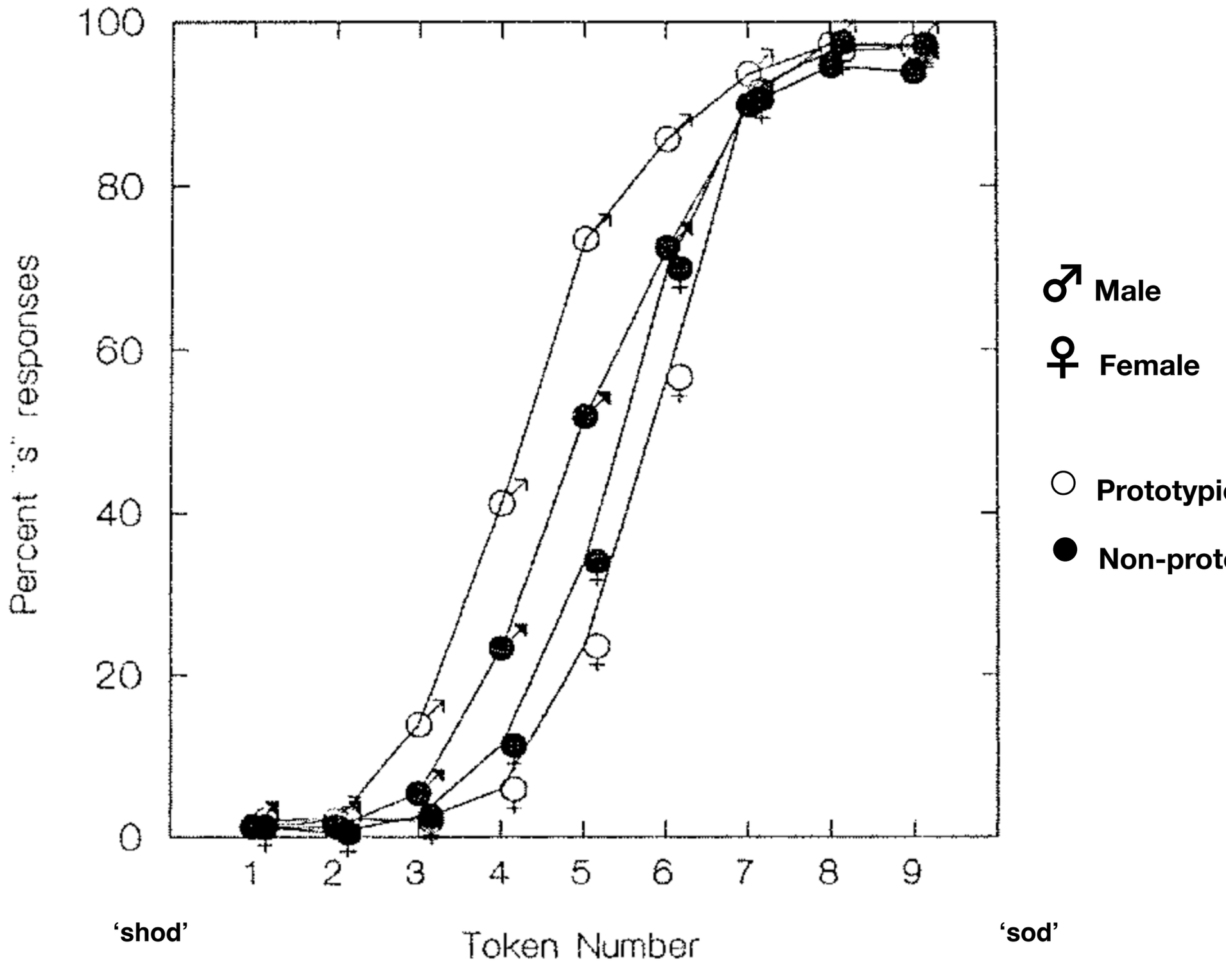
- a single linguistic signal (text, recording, speaker)
- multiple social categories (e.g. photos, videos, stuffed animals)
- elicit a segmental or lexical response, often with reaction time or eye tracking behavioral measures

Gradient and Visual Speaker Normalization in the Perception of Fricatives
Strand & Johnson (1996)

- Within-speaker, [ʃ] has a lower center of gravity (amplitude-weighted average frequency) than [s]
- Listeners report perceiving a larger proportion of an [ʃ]-[s] continuum as [s] when the vowels were from a larger vocal tract, more [ʃ] from a smaller vocal tract (May 1976; Mann & Repp 1980)

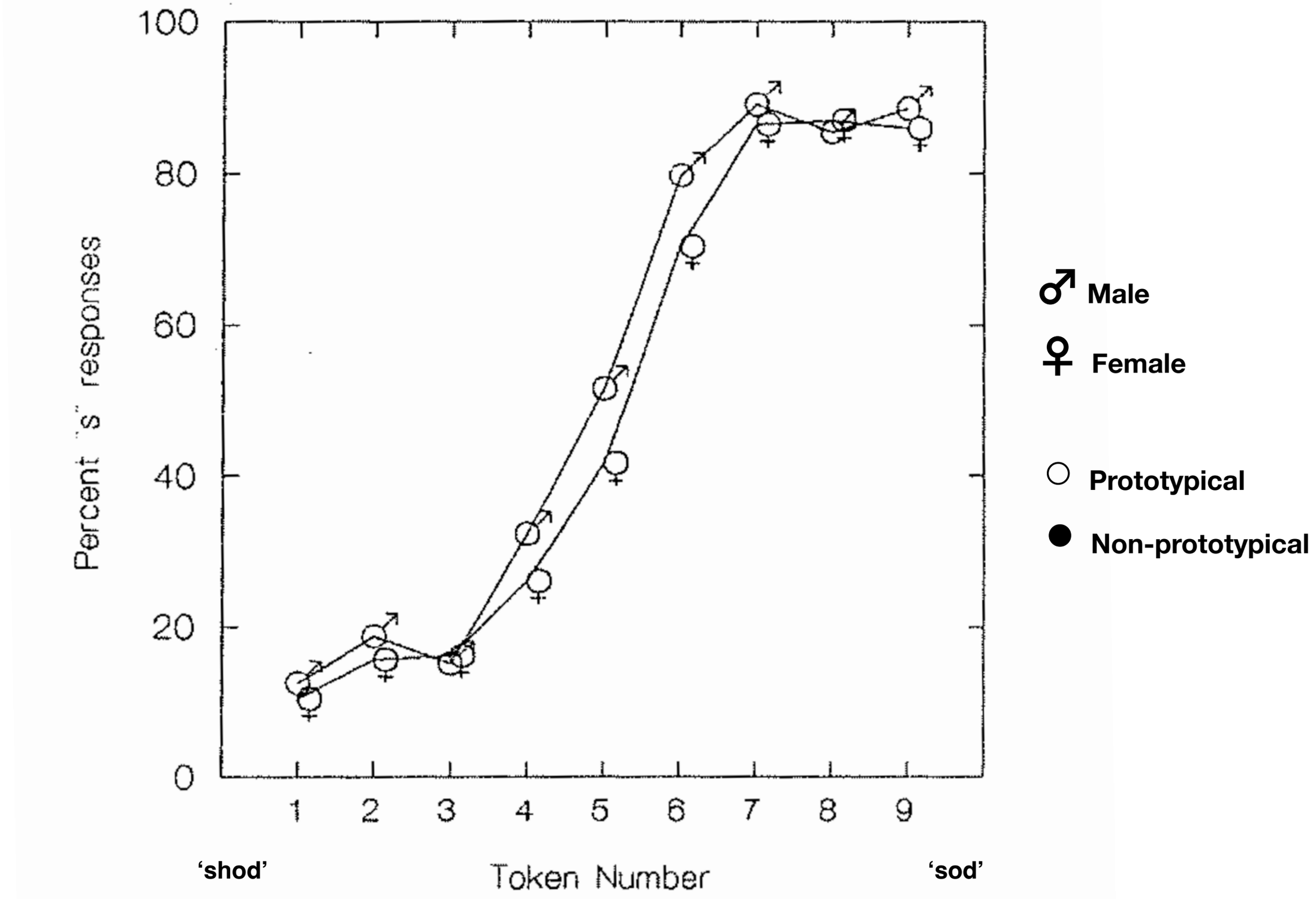
Listeners' [ʃ]-[s] category boundary depends on the perceived gender of the talker's voice

Strand & Johnson (1996): Experiment 1



Listeners' [ʃ]-[s] boundaries for non-prototypical male voice shifted by guise presentation

Strand & Johnson (1996): Experiment 2



Strand & Johnson (1996)

Relevant Findings:

- **Experiment 1**: Voices carry social information, and gradient social information can have a gradient influence on fricative identification
- **Experiment 2**: Strand Effect: the same token was perceived differently when paired with a male face than when paired with a female face

A recent Strand Effect replication:

- Bouavichith, Calloway, Craft, Hildebrandt, Tobin, and Beddor (2019) replicated and extended the Strand Effect using eye tracking
- **Visual Guise**: Listeners saw one of these two faces from the Chicago Face Database:



Bouavichith et al. (2019)

- Participants (exp1) were asked to look at the word they heard along a synthetic *sack*-*shack* continuum





Bouavichith et al. (2019)

- Gender typicality of the voice was also manipulated in a 5 step F0 and Formant Ratio continuum at each continuum step





The Strand Effect is bi-directional

- Listeners seeing a prototypical male face heard more of the continuum as [s], listeners seeing a prototypical female face heard more of the continuum as [ʃ] (replicating Strand & Johnson)
- and!!!!
- Listeners asked to decide who was talking showed the pattern in reverse; listeners presented with [ʃ]-like tokens in otherwise gender-ambiguous tokens looked more to the female face while listeners presented with [s]-like tokens looked more to the male face

Matched Guise & Awareness

Assumption: the connection between speech and social category requires a lack of conscious awareness; Unlike McGurk Effect, requires participant to believe the guise

Experimenters take great pains to hide the guise manipulation (e.g. Grondelaers & van Gent 2019; Pharao & Kristiansen 2019) or assume that any behavioral change means the guise was believed (e.g. Rubin 1992, Lippi-Green 1997; 2011)

Q1: Do listeners *have to* believe the guise?

Q2: If the Strand Effect is bi-directional, what happens if face and voice disagree?

Revealing the Guise

- Replicate and extend the Strand Effect
- Using Bouavichith et al.'s stimuli (faces, fricative continuum, vowels, everything), we created a new pair of Instruction conditions:
  - **Hidden**: Bouavichith et al. (2019) exp1 replicated in a web browser, normal matched guise instructions
  - **Unhidden**: participants were instructed "the face you see is not the voice you are listening to…"

Face x Voice

- In addition to **Instruction** we also manipulated **Congruity**
- We paired each face with continua generated using the prototypically female talker's voice -and- continua generated using the prototypically male talker's voice
- This created **Congruous** (Male face/Male source voice; Female face/Female source voice) Strand & Johnson exp2 style pairings and **Incongruous** (Male face/ Female source voice; Female face/Male source voice) pairings

Congruity: Face x Voice

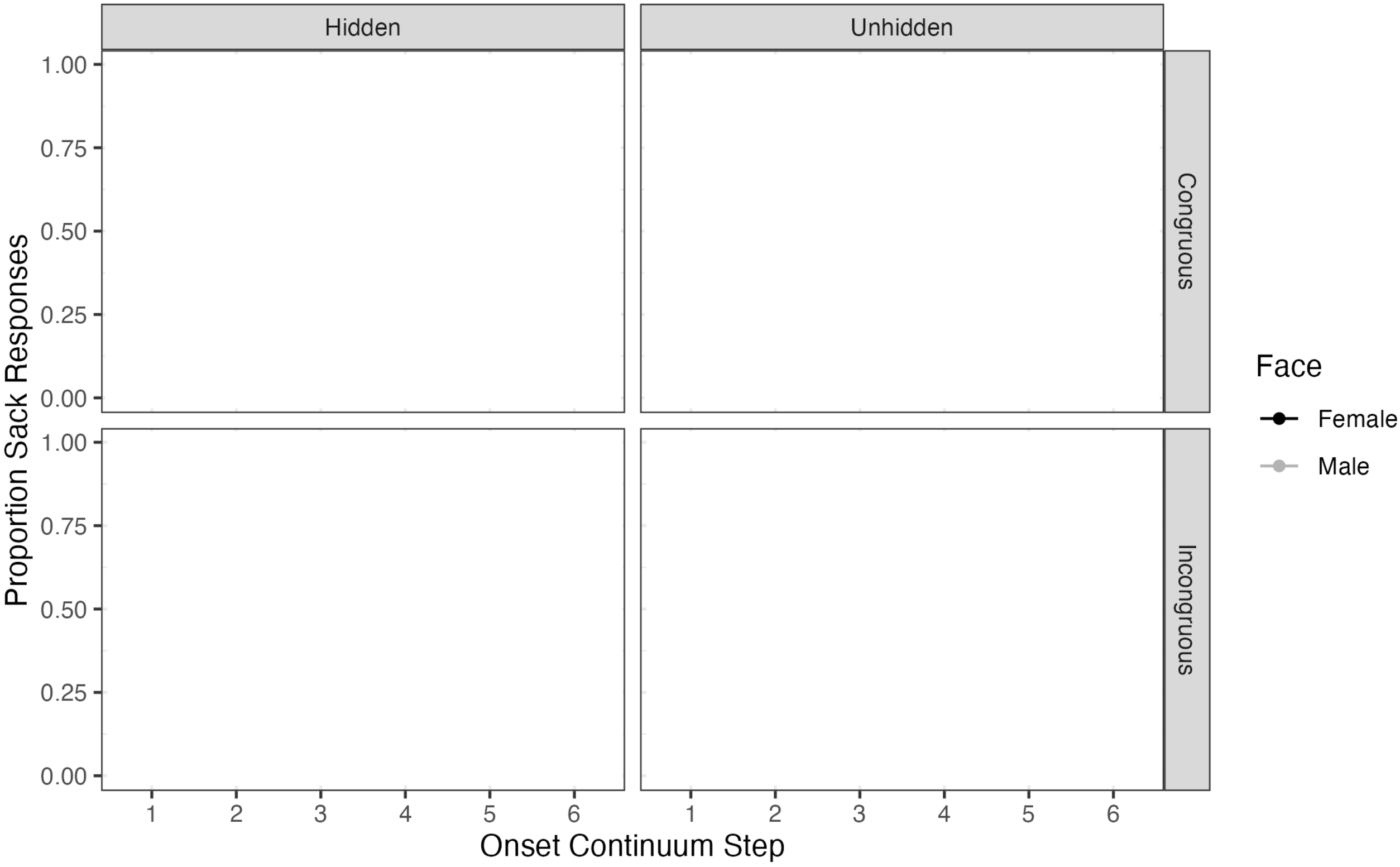
- We also manipulated **Congruity** of the gender identity of the stimulus talkers and the expectation created by the images

	Female Voice	Male Voice
 Female Face		
 Male Face		

Laycock & McGowan

- 120 Total Participants
    - 30 Hidden + Congruous
    - 30 Unhidden + Congruous
    - 30 Hidden + Incongruous
    - 30 Unhidden + Incongruous
- Recruited using Prolific.com
- Experiment written in OpenSesame/JATOS
- 14,400 total trials

Results: Instructions



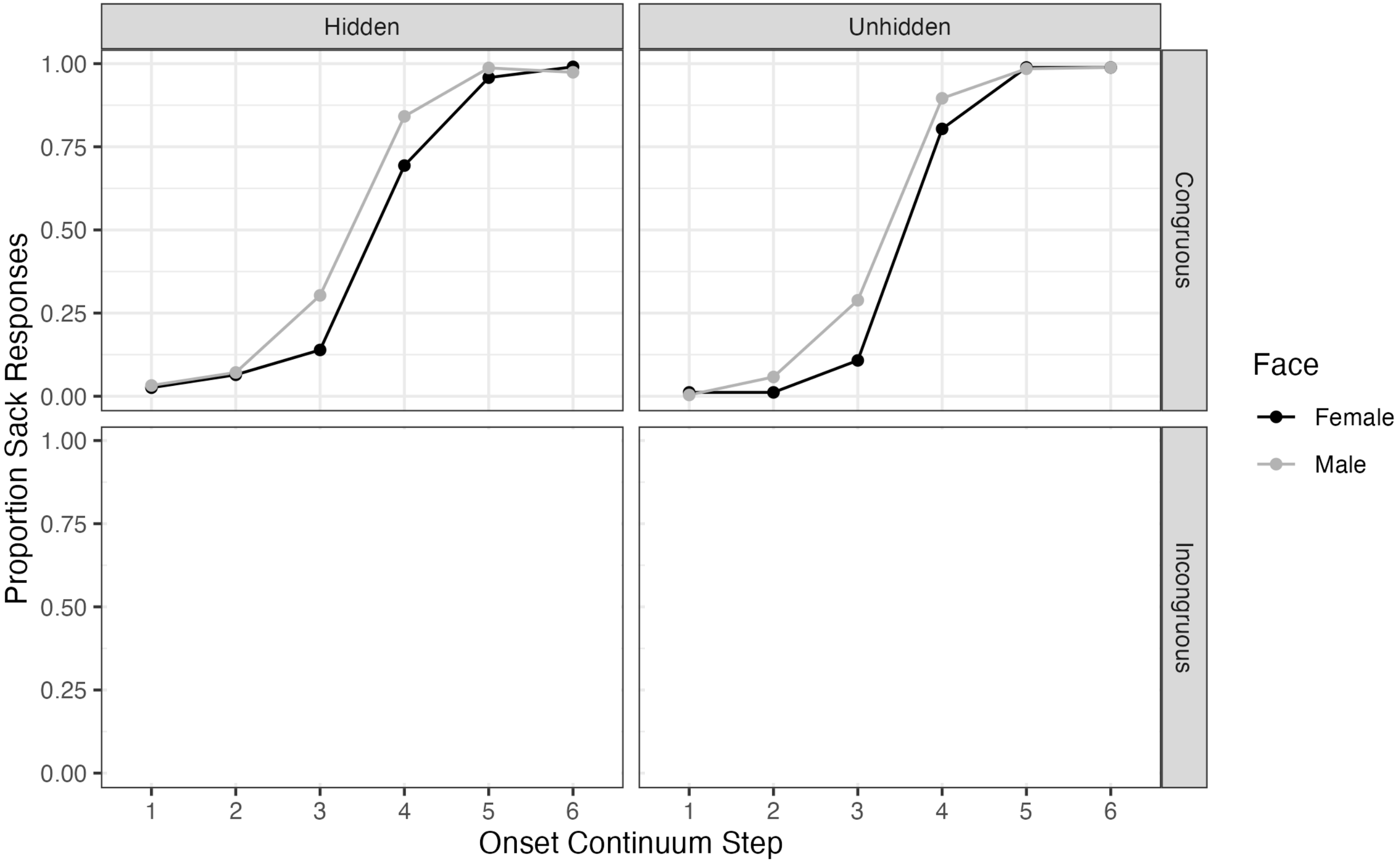
Discussion: Instruction

- Revealing the nature of the guise manipulation did not significantly influence listener responses
- Social knowledge influences perception even when listeners are aware that it is, or may be, false
  - This is inconsistent with a model in which that social knowledge of speech is available for conscious scrutiny or uniquely delicate in the face of listener awareness (e.g. Labov et al. 2011)

Discussion: Instruction

- Campbell-Kibler (2020), similar finding with language attitudes. In three conditions:
    - Control (participants believe face and voice match)
    - Voice (participants instructed to disregard face)
    - Face (participants instructed to disregard voice)
- Accentedness and Attractiveness ratings revealed some ability to ignore social information when instructed to do so
- However listeners in Campbell-Kibler (2020) found social information from the voice more difficult to ignore…

Results: Congruity



Discussion: Congruity

- The Strand Effect disappears when gender information in the voice and gender information in the face conflict
- Listeners still attribute more of the continuum to *sack* [s], but that pattern follows the voice, not the face.
- As in Campbell-Kibler (2020), McGowan and Babel (2020), McGowan (2015), Rubin (1992), etc. our participants found social information in the voice difficult to ignore in the presence of conflicting visual information
    - This holds both when listeners are aware the visual social information is, or could be, false and when instructed otherwise.

Conclusions

- Believing that the face and voice are from the same person does not appear to matter in either type of Matched Guise
- Voices carry at least as much, probably more, social information than the other sources we use in these experiments (not unlike Whalen 1984?)
- Taken together, these results suggest that we are worrying about the wrong thing when focusing on believability of matched guise tasks when we should be focused on understanding all the potential sources of information in our stimuli

Boyd, Fruehwald, and Hall-Lew (2021)

Crosslinguistic Perceptions of /s/

- English listeners associate fronted [s+] and Gayness regardless of language, regardless of knowledge of language
- French and German listeners do *not* associate fronted [s+] with Gayness for any language (including their own) even though this feature is present in the natural speech of gay men in both French and German (Boyd 2018a, 2018b)

References

Bouavichith, D. A., Calloway, I. C., Craft, J. T., Hildebrandt, T., Tobin, S. J., & Beddor, P. S. (2019). Bidirectional effects of priming in speech perception: Social-to-lexical and lexical-to-social. The Journal of the Acoustical Society of America, 145. doi: 10.1121/1.5101933

Boyd, Z. (2018a). Crosslinguistic variation of =s= as an index of non-normative sexual orientation and masculinity in French and German men. Doctoral dissertation, The University of Edinburgh.

Boyd, Z. (2018b). A certain kind of gay identity: [sþ] and contextually mediated variation in bilingual French and German men. Unpublished Manuscript.

Boyd, Z, Fruehwald, J., & Hall-Lew, L. (2021). Crosslinguistic perceptions of /s/ among English, French, and German listeners. Language Variation and Change, 33(2), 165–191.

Campbell-Kibler, K. (2009). The nature of sociolinguistic perception. Language Variation and Change, 21(1), 135-156.

Kircher, R. (2015). The matched-guise technique. Research methods in intercultural communication: A practical guide, 196-211.

Labov, W., Ash, S., Ravindranath, M., Weldon, T., Baranowski, M., & Nagy, N. (2011). Properties of the sociolinguistic monitor. Journal of Sociolinguistics, 15(4), 431–463.

References

Lambert, W. E., Hodgson, R. C., Gardner, R. C., & Fillenbaum, S. (1960). Evaluational reactions to spoken languages. The journal of abnormal and social psychology, 60(1), 44.

Lippi-Green, R. (1997). English with an accent: Language, ideology, and discrimination in the United States. London, UK: Routledge.

McGowan, K. B. (2015). Social expectation improves speech perception in noise. Language and Speech, 58(4), 502–521.

McGowan, K. B. & Babel, A. M. (2020). Perceiving isn't believing: Divergence in levels of sociolinguistic awareness. Language in Society, 49(2), 231–256.

McGurk, H., & MacDonald, J. (1976). Hearing lips and seeing voices. Nature, 264(5588), 746-748.

Rubin, D. L. (1992). Nonlanguage factors affecting undergraduates' judgments of nonnative English-speaking teaching assistants. Research in Higher Education, 33(4), 511–531.

Strand, E. A., & Johnson, K. (1996). Gradient and visual speaker normalization in the perception of fricatives. In Konvens (pp. 14–26).

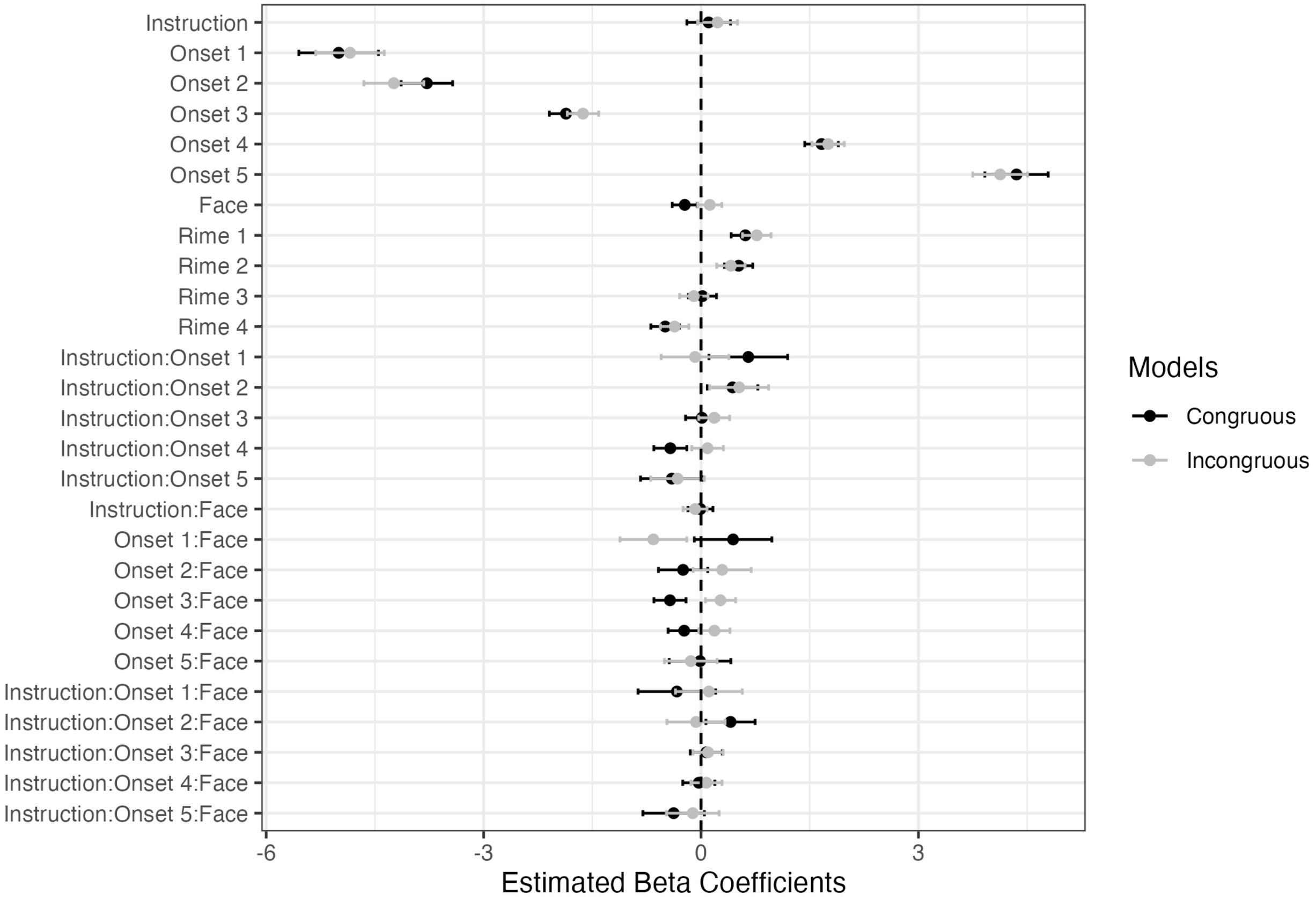
Whalen, D. H. (1984). Subcategorical phonetic mismatches slow phonetic judgments. Perception and Psychophysics, 35, 49-64.

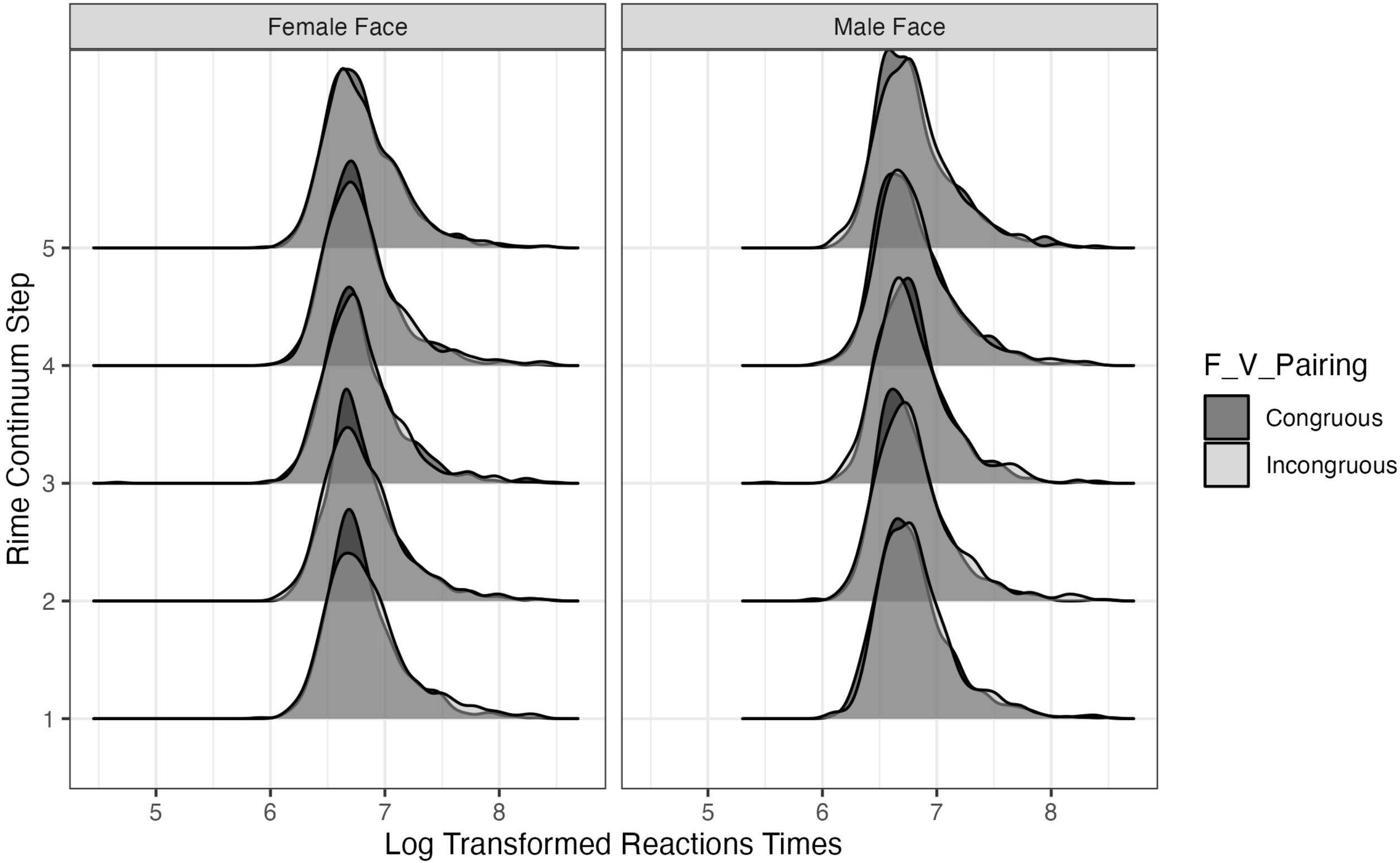
Thank You!

Kyler Laycock
laycock.21@osu.edu

Kevin B. McGowan
kbmcgowan@uky.edu

MidPhon 28
Purdue University
21 October 2023


Instruction
Onset 1
Onset 2
Onset 3
Onset 4
Onset 5
Face
Rime 1
Rime 2
Rime 3
Rime 4
Instruction:Onset 1
Instruction:Onset 2
Instruction:Onset 3
Instruction:Onset 4
Instruction:Onset 5
Instruction:Face
Onset 1:Face
Onset 2:Face
Onset 3:Face
Onset 4:Face
Onset 5:Face
Instruction:Onset 1:Face
Instruction:Onset 2:Face
Instruction:Onset 3:Face
Instruction:Onset 4:Face
Instruction:Onset 5:Face
-6
-3
0
3
Estimated Beta Coefficients
Models
Congruous
Incongruous


Female Face
Male Face
Rime Continuum Step
5
4
3
2
1
5
6
7
8
Log Transformed Reactions Times
F_V_Pairing
Congruous
Incongruous

Hidden
Unhidden
Congruous
Incongruous
Proportion Sack (Onset steps 3 & 4)
1.00
0.75
0.50
0.25
0.00
1
2
3
4
5
Rime Continuum Step (from Male = 1 to Female = 5)
Speaker
Male
Female